# Zekun Wang

zekunw@kth.se

My recent research interest is federated learning in the context of cloud computing and IoT while having a background of network security. I'm looking for chances to gain insight into industry and frontier technology by proposing, designing and implementing innovative FL applications powered by networking technologies, optimization theory, etc.

## Education

**University of Tokyo (UTokyo)**, ***Tokyo, Japan*** ***09/2022-04/2023***
*Exchange Study* *Major: Creative Informatics* *GPA: 4.0 / 4.0*

- ***Relevant Coursework***: Network Optimizations, Parallel and Distributed Programming
- ***Honors***: JASSO Scholarship

**KTH Royal Institute of Technology (KTH)**, ***Stockholm, Sweden*** ***09/2021-Present***
*Master of Science* *Major: Communication Systems* *GPA: 4.81 / 5.0*

- ***Relevant Coursework***: Advanced Internetworking, Mobile Networks and Services, SDN & NFV
- ***Honors***: KTH Scholarship

**Huazhong University of Science and Technology (HUST)**, ***Wuhan, China*** ***09/2018-06/2022***
*Bachelor of Engineering* *Major: Electronic and Information Engineering* *GPA: 3.89 / 4.0*

- ***Relevant Coursework***: Stochastic Processes, Computer Networks, Principles and Applications of Big Data Technology
- ***Honors***: Outstanding Undergraduates in Term of Academic Performance (awarded to top 1%), Scholarship for Academic Excellence, Excellent Student Association Cadre

## Research Experience

**Confidential Federated Learning with Homomorphic Encryption**, ***Stockholm, Sweden*** ***03/2023-09/2023***
*Master Thesis in CanaryBit* *supervised by Dr. Nicolae Paladi*

- Built a privacy-preserving FL framework enhanced by several homomorphic encryption schemes and trusted execution environment, protecting both data confidentiality and integrity during the whole life-cycle.
- Designed an optimized HE scheme specifically for FL, achieving higher flexibility and low traffic overhead and significantly mitigating the risk of overflowing when aggregating.
- Tested the proposed framework on a threat intelligence dataset and conducted a comparison of supported HE schemes in various settings.

**Research on Applying Multi-access Edge Computing & Federated Learning in IoV Scenario** , ***Tokyo, Japan*** ***11/2022-05/2023***
*Research Project in Tsukada Lab* *supervised by Prof. Manabu Tsukada*

- Proposed a vehicle-focused FL framework called Flames that enhances task-specific model caching across various conditions.
- Designed mechanisms to manage cached models and control FL processes to reduce memory overhead and increase efficiency while retaining model performances.
- Evaluated it on BDD100K dataset, achieving a 5% increase of overall accuracy against centralized training.

**Intelligent Reflecting Surfaces for Improved Coverage**, ***Stockholm, Sweden*** ***12/2021-01/2022***
*IK2560 Course Project in KTH* *supervised by Prof. Emil Björnson*

- Investigated several optimization algorithms and implemented BPSO to optimize IRS configuration for maximum data rate.
- Developed a variant of BPSO called Loop-BPSO to address the observed shortcomings of BPSO, slightly accelerating the searching speed and obtaining a higher score.

## Project Experience

**Summer Internship in Ericsson**, ***Kista, Sweden*** ***06/2022-09/2022***
*Analyzing and Visualizing the Performance of Distributed Bazel Compiling System*

- Developed the Bazel Event Service Application in Java, which communicates with multiple Bazel clients, processes and sends the received data to the OpenSearch backend. Use OpenSearch for data storing, large data analysis and visualization.
- Modified the source code of Bazel to enhance the Bazel Event Protocol, making it possible to gain comprehensive client information.
- Followed an Agile workflow, and implemented many cooperative developing methods to increase the team efficiency.

## Skills

| | |
|---|---|
| **Programming** | Python, C/C++, Java, Matlab, Javascript |
| **Language** | English (Professional working proficiency), Chinese (Native), Japanese (Visitor Level) |